Ho Huu Binh

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Education

University of Science – Vietnam National University Ho Chi Minh City, Vietnam
M.S. in Mathematical Statistics (Dec 2025)

- *Thesis:* Model-based Clustering with Variable Selection under MNAR Data
- *Supervisors:* Dr. Hoang Van Ha (HCMUS), Dr. Nguyen Trung Tin (QUT)
- *Honors:* Odon Vallet Scholarship (2025) - a nationally competitive award recognizing outstanding academic performance and research potential.

International University – Vietnam National University Ho Chi Minh City, Vietnam
B.S. in Applied Mathematics *Aug 2018 – Jul 2022*

- *Thesis:* Forecasting Unit Sales of Retail Goods using Dynamic GLMs (DGLMs)
- *Supervisor:* Dr. Pham Hai Ha (HCMIU)
- *Honors:* First Prize, Vietnam National Olympic Econometrics Contest (2021); Third Prize, Scientific Conference for Students (2021)

Research Interests

- **Mathematical Statistics:** High-dimensional statistics, mixture models, robustness under model misspecification.
- **Bayesian Inference:** Variational Bayes, Bayesian nonparametrics, uncertainty quantification.
- **Statistical Machine Learning:** Mixture-of-Experts architectures, generative modeling, missing-data mechanisms (MAR/MNAR), representation learning.
- **Time Series and Stochastic Models:** Continuous-time modeling, state-space models, probabilistic forecasting.

Publications & Preprints

[1] **Binh H. Ho**, Long Nguyen Chi, TrungTin Nguyen, Binh T. Nguyen, Van Ha Hoang, Christopher Drovandi - *A Unified Framework for Variable Selection in Model-Based Clustering with Missing Not at Random.* arXiv:2505.19093v1 [stat.ML]. (Accepted at NeurIPS 2025)

[2] Ta, B. Q., Huynh, V. T., Nguyen, K. Q. H., Nguyen, P. N., & **Ho, B. H.** (2022). *Maximal predictability portfolio optimization model and applications to Vietnam stock market.* In: Studies in Systems, Decision and Control. Springer. (Peer-reviewed Conference Proceedings, TES 2022)

Research Experience

Revisiting Parameter Balancing in Kinetic Models of Cell Metabolism (Ongoing)

- Constructing a robust Bayesian framework for estimating kinetic parameters in metabolic networks under model misspecification and limited observational data.
- Employing hierarchical Student's-t structures to handle heavy-tailed observational noise, infer latent metabolic fluxes, and provide principled uncertainty quantification.
- Aiming to offer a statistically grounded alternative to traditional parameter-balancing procedures in systems biology, with emphasis on robustness and stability.

Dynamical Signature Network for Time Series Classification (Ongoing)

- Proposing a continuous-time representation learning approach based on path signatures and MoEs to extract stable and expressive features for irregular time series.
- Studying model stability through Lipschitz control of the signature map and Jacobian-based regularization, with the goal of establishing theoretical generalization guarantees.
- Utilizing Random Fourier Features to approximate kernelized signature representations and implementing the method on benchmark sequence datasets.

Gated-MNAR Mixture Model for High-Dimensional Clustering (Ongoing)

- Developing a mixture-of-experts framework that naturally incorporates Missing Not At Random (MNAR) mechanisms by linking cluster assignments with a gating network describing missingness patterns.
- Analyzing identifiability conditions and implementing an EM-type algorithm adapted to MNAR settings and evaluating its performance on synthetic and biomedical datasets with structured missingness.

Deep Learning for Financial Risk and Portfolio Optimization

- Constructed a NeuralNet–GARCH model to estimate conditional volatilities and forecast Value-at-Risk, observing significant improvements over ARIMA–GARCH baselines.
- Designed a portfolio allocation model incorporating attention-based feature weighting, yielding improved risk-adjusted performance relative to classical benchmarks.
- Formulated and optimized a non-convex fractional quadratic program to analyze predictability in asset returns, comparing results with the standard mean–variance framework.

Industry Experience

Manulife Ho Chi Minh City, Vietnam
Product Development Intern *May 2022 – Nov 2022*

- Analyzed insurance product cash-flow structures and implemented statistical procedures in **R** to support premium rate derivation for critical-illness products.
- Automated data extraction from internal data lakes using **SQL**, assisting in product validation and market comparison studies.

FPT Japan – Usee Ho Chi Minh City, Vietnam
Inventory Optimization and Forecasting Intern *May 2021 – Nov 2021*

- Implemented probabilistic and machine-learning forecasting models to improve sales prediction accuracy for intermittent demand with zero-inflated patterns.
- Evaluated inventory management strategies for pharmaceutical products, focusing on balancing stockout risk and holding costs through statistical modeling.

Technical Skills

- **Programming:** Julia, Python, R (including Rcpp), SQL
- **Statistical and ML Frameworks:** PyTorch, JAX, NumPyro, Pyro, PyMC3, Scikit-learn, Nixtla, GluonTS
- **Bayesian Inference Tools:** NumPyro, Pyro, PyMC3
- **Optimization and OR Tools:** GUROBI, CVXPy, OR-Tools, PuLP

References

- **Dr. Hoang Van Ha**
  Lecturer, Faculty of Mathematics and Computer Science
  University of Science, VNU-HCMC
  Email: hvha@hcmus.edu.vn

- **Dr. Nguyen Trung Tin**
  Lecturer, School of Mathematical Sciences
  Queensland University of Technology (QUT)
  Email: t600.nguyen@qut.edu.au

- **Dr. To Duc Khanh**
  Lecturer, Faculty of Mathematics and Computer Science
  University of Science, VNU-HCMC
  Email: tdkhanh@hcmus.edu.vn